

Qun Li

Ericli6897@gmail.com | LinkedIn | GitHub | Website

TECHNICAL SKILLS

Languages: Python, Go, C++, Ruby, JavaScript, TypeScript, Java, SQL (PostgreSQL)

Frameworks & Platforms: FastAPI, Ruby on Rails, React, Node.js, LangChain, LangGraph, Spring AI, REST API

Developer Tools & Infrastructure: Linux, Git, Docker, Jenkins, Redis, PostgreSQL

Data & AI Libraries: Pandas, NumPy, Scikit-learn, XGBoost, Optuna

Concepts: Software Engineering, System Design, Full-Stack, Backend, Frontend, Cloud Infrastructure, Data Structures and Algorithms, CI/CD Pipeline, Agile Methodologies, Web Development, Database, HTTP Protocols, Product Design

PROJECTS

UofT Dating — *React, TypeScript, Python, PostgreSQL, WebSocket* Jan. 2026

- Engineered a **exclusive** dating website for the University of Toronto re-architecting open-source framework into a closed system that enables verified students to discover, search, and communicate with peers in a trusted campus environment.
- The platform features OTP-verified university email authentication, a UofT-branded React frontend using **TypeScript**, PostgreSQL persistence, **WebSocket** messaging, and Q&A-driven matching logic connecting users via response similarity

SQL AI Agent — *FastAPI, LangChain, PostgreSQL, SQLAlchemy, Docker* Dec. 2025

- Built a **ChatGPT**-style assistant for SQL databases where users ask questions in English, the system generates executable SQL, queries PostgreSQL, and returns both an answer and raw SQL results for verification and debugging
- Implemented a **FastAPI REST service** with persistent chat memory, packaged PostgreSQL in Docker so setup stays consistent and runs remain reproducible for local development and testing across different environments

API Automation Testing — *FastAPI, Docker, Jenkins* Nov. 2025

- Built and led a **CI-driven** FastAPI testing platform integrated with Docker and Jenkins, running Excel-defined API test cases per build, capturing outcomes, and generating HTML/Excel reports for review after each pipeline run
- Implemented a regression workflow mapping business scenarios to **API calls** and expected status codes, helping validate backend changes faster, catch breaking updates earlier, and reduce manual retesting iterations

House Price Prediction — *XGBoost, Optuna, Scikit-Learn, FastAPI, Docker* Oct. 2025

- Built an end-to-end **ML pipeline** using XGBoost with Optuna tuning inside a scikit-learn workflow, improving performance through preprocessing and model selection, and ranking **Top 2%** among 5,000+ Kaggle participants
- Deployed the model as a FastAPI REST service, containerized with **Docker** and shipped through Render using GitHub auto-deploy, making releases repeatable and enabling updates when the pipeline or model version changes

EXPERIENCE

Math Instructor (Part-Time) Sep. 2025 – Oct. 2025
Kumon Learning Centre, Toronto

- Provided 1:1 tutoring on multi-step math questions by modeling full solutions, checking student work line-by-line, and explaining why each step works, with extra practice on algebra, fractions, and word problems to build consistency
- Reviewed and graded daily assignments to identify recurring error patterns, track learning gaps across topics, and adjust pacing with targeted worksheets, ensuring students practiced the exact skills they struggled with most

Math Tutor Dec. 2024 – Apr. 2025
Self, Toronto

- Tutored students weekly on Euclid contest material including algebra, geometry, and number theory, creating focused practice plans and teaching reusable solution patterns that students could apply confidently to unfamiliar questions
- Designed mock tests and feedback cycles to improve speed and accuracy, reviewing mistakes by category and explaining cleaner approaches, helping all students reach Top 25% Distinction in the Euclid Contest across Canada

EDUCATION

University of Toronto Toronto, Canada
Bachelor of Science in Applied Statistics and Applied Mathematics Expected Jun. 2027

- Year GPA: 3.3/4.0. Relevant Coursework: Data Structures and Analysis, Systems Programming, Software Design, Databases, Algorithm Design, Machine Learning, Artificial Intelligence, Software Tools and Testing